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Modelling of MSRE graphite temperature in porous-medium multi-physics simulations

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ABSTRACT

MSRs stand out as prominent candidates among advanced reactor designs, addressing the global demand for safer and more sustainable nuclear energy. Accurate multi-physics modelling is essential for the advancement of MSR technology, particularly for understanding the thermos-hydraulic behaviour of graphite under irradiation. This study focuses on developing and implementing a high-fidelity methodology within the GeN-Foam code to model graphite temperature distribution within porous-medium multi-physics simulations, using the MSRE as a benchmark. The approach combines thermal-hydraulic and neutronic modelling by using Serpent-generated cross-section data as inputs for the **Gen-Foam** neutronic solver. Validation against MSRE measurements performed at ORNL benchmark data confirms the framework's reliability. The axial temperature distribution yields a Mean Absolute Percentage Error (MAPE) of 1.09%, while the radial distribution shows a MAPE of 0.62%. The average graphite temperature of 935.6 K is consistent with the ORNL reference value of 936.4 K under steady-state conditions.

1. Introduction

Molten Salt Reactors (MSRs) are among the leading advanced reactor concepts being considered for the future of nuclear energy. They offer multiple advantages, including fuel-cycle flexibility, intrinsic safety features, reduced environmental impact, and the potential for actinide burning (Fiorina et al., 2013), and suitability for a range of possible applications, such as producing high-temperature process heat for industrial applications, desalination of water, and as a power source for remote locations (DoE, 2002; de Figueiredo Luiz et al., 2024; Forsberg, 2007; Park et al., 2024; Serp et al., 2014).

In the 1960s, Oak Ridge National Laboratory (ORNL) conducted several studies to identify the best MSR design for civilian power generation. This research concluded that a graphite-moderated MSR was the most promising option (Rosenthal et al., 1970). The Molten Salt Reactor Experiment (MSRE), a fluid-fuelled test reactor, successfully operated at ORNL from 1966 to 1969, and remains the most significant experimental reference for graphite-moderated MSRs (Stewart, 2020). The MSRE used a fluoride-based fuel salt circulating through channels formed by prismatic graphite blocks, which served both as moderator and structural material. Most of the fission energy was deposited in the circulating salt, while 6% of the reactor power is produced in the graphite in the absence of fuel permeation. This value is based on calculations of gamma and neutron heating in the graphite (Engel and Haubenreich, 1962).

Unlike conventional solid-fuel reactors, MSRs utilise a circulating liquid fuel, leading to strongly coupled physical phenomena. The tight interaction between neutronics, thermal-hydraulics, and fuel transport processes requires a multi-physics modelling approach to accurately simulate the reactor's behaviour.

In recent decades, several multi-physics simulation tools have been developed to simulate graphite-moderated, channel-type MSRs (Fiorina et al., 2015; Krepel et al., 2005; Křepel et al., 2007; Li and Cai, 2018). Among these efforts, the OpenFOAM® framework has also been adopted for MSR applications (Aufiero et al., 2014; Hu et al., 2017; Radman et al., 2021)due to its flexibility in handling complex geometries and coupled physics. For the same reasons, OpenFOAM has seen growing use in the analysis of other advanced reactor systems as well. One such OpenFOAM-based platform is GeN-Foam (Fiorina et al., 2015, 2016; Radman et al., 2021), developed through collaboration between the Paul Scherrer Institute (PSI) and EPFL's Laboratory for Reactor Physics and Systems Behaviour (LRS). GeN-Foam is specifically designed to simulate 3D thermal-hydraulics (with both CFD and porous-medium

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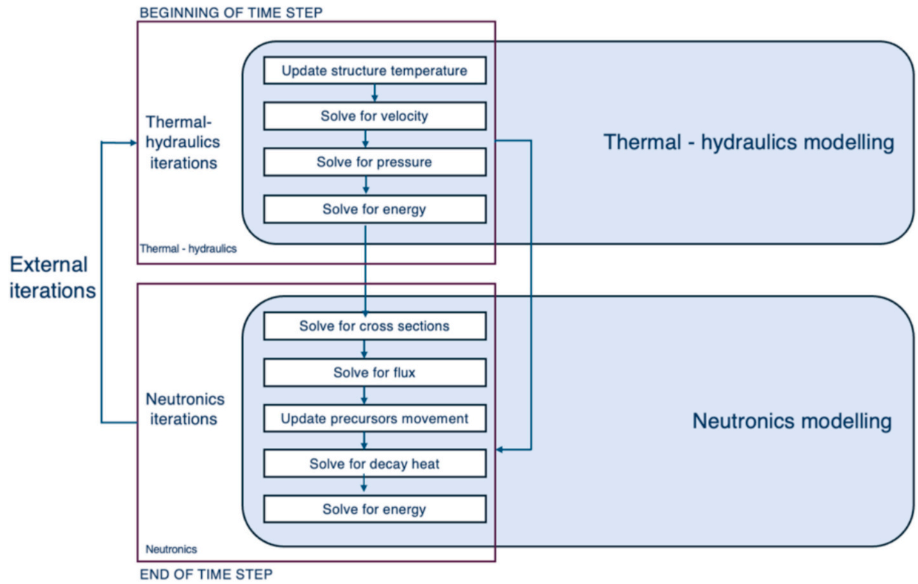

Fig. 1. Gen-Foam multi-physics workflow model for studying MSRE graphite temperature.

Table 1
MSRE design and operation parameters (Bao, 2016).

	Value
Design full power	10 MWth
Actual full power[1]	~7.4 MWth
Hot leg temperature	654.4°C (1210 °F) (927.6 K)
Cold leg temperature	632.2 °C (1170 °F) (905.4 K)
Core ΔT	22.2 °C (40 °F)
Primary flow rate	0.07571 m^3/s (1200 gpm)
Primary pressure	0.03447 MPa (5 psig)
Primary fuel salt	LiF-BeF2-ZrF4-UF4
Fuel melting point	434 °C (813 °F)(707.0 K)
Primary salt density	2241 kg/m^3 (139.9 lb./ft^3) At 650 °C
Moderator	Graphite

[1] The MSRE was designed for a maximum thermal power of 10 MW, but it operated at much lower power levels because its mission was experimental rather than power-producing. Most experiments required only a modest neutron flux, and operating well below the design power provided larger safety margins and simplified heat removal. For these reasons, the actual operating power of the MSRE was intentionally kept far below the design value.

approaches), neutronics, and thermo-mechanics using unstructured and deformable meshes (Fiorina et al., 2015). Although its use for MSRs has been limited to date (Bao, 2016), its modular structure and multi-physics capabilities make it suited for the development of high-fidelity MSR simulation tools. In this study, GeN-Foam Master version built on OpenFOAM version 2312 is used.

This work presents the preliminary development and validation of a multi-physics model for thermal, graphite-moderated MSR using the GeN-Foam platform. While the **ultimate goal** of this project is to study **graphite behaviour** under irradiation and prolonged operation, as well as its effects and interactions with other physics, this study focuses on validating **the multi-physics** model against experimental data from the MSRE. Establishing this validation will provide a solid foundation for future studies aimed at **characterising graphite behaviour** in MSRs.

This paper is structured as follows: Section 2 outlines the modelling framework, including the reactor description, the numerical tools and approaches, geometry and meshing, as well as the physical models and boundary conditions employed. Section 3 presents the results and discussion, covering both neutronic and thermal-hydraulic analyses and comparisons with experimental data. Finally, Section 4 concludes with a summary of the key findings and their implications.

2. Modelling framework

This section outlines the multi-physics modelling framework developed for simulating the MSRE, highlighted in Fig. 1. First, it provides an overview of the key design features of the MSRE, then it describes the numerical approach and main modelling assumptions, as well as the mesh and physical models used in this study.

2.1. MSRE description and benchmark basis

In this study, experimental data from several ORNL reports are utilised for modelling and validation. These include:

- ORNL-TM-3229, *Fluid dynamic studies of the molten-salt reactor experiment (MSRE) core,* focusing on describing the models, the testing, and the data from which velocity, pressure drop and flow patterns, also describing how measurements were extrapolated to molten salt at 922 K in the actual reactor (Kedl, 1970);
- ORNL-TM-0378, *Temperature in the MSRE core during steady-state power operation*, focusing on the temperature distribution in different parts of the reactor and effects of power on the nuclear mean temperatures that were combined with the temperature coefficients of reactivity of the fuel and graphite to estimate the power coefficient of reactivity of the reactor (Engel and Haubenreich, 1962);
- ORNL-TM-730, *MSRE design and operation report part III*, focusing on considerations of the effects of core size and fuel-to-moderator ratio on critical mass and fuel concentration, the nuclear characteristics of the reactor, both the fuel and the moderator temperature coefficients of reactivity, reactivity coefficients and temperature distributions in the fuel and graphite in the reactor for estimating power level (Haubenreich et al., 1964);
- ORNL-TM-251, *Safety calculations for MSRE*, focusing on the conceivable reactivity accidents that were analysed, using conservative assumptions and approximations, to permit evaluation of reactor safety. None of the accidents which were analysed lead to catastrophic failure of the reactor, which is the primary consideration. Some internal damage to the reactor from undesirably high temperatures could result from extreme cold-slug accidents, premature criticality during filling, or uncontrolled rod withdrawal (Haubenreich and Engel, 1962);

Table 2
Properties of the MSRE fuel salt at 922 K (De Wet and Greenwood, 2019).

Parameter	Value
Specific heat	2×10^3 J/(kg .K)
Thermal conductivity	5.5 W/(m.K)
Dynamic viscosity	8.06×10^{-3} kg/(m .s)
Reactor initial fuel temperature	722 K

Table 3
MSRE thermo-physical properties (Forsberg, 2007; Mateusz Pater, 2019).

Parameter	Value
Graphite Density	1874 kg. m^{-3}
Graphite thermal conductivity	53 $Wm^{-1}k^{-1}$
Graphite heat capacity	1772 J $Kg^{-1}k^{-1}$

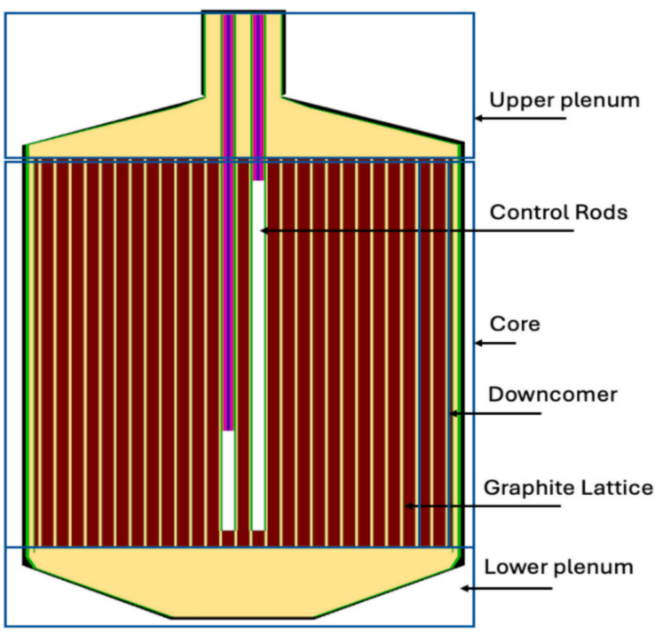

Fig. 2. Zone layout for the components inside the vessel.

- ORNL-TM-728, "MSRE design and operation report part I", focusing on the description of reactor design and general description, site, plant, fuel circulation system, coolant salt circulation system (Robertson, 1965) and
- ORNL-TM-2098, *Tube vibration in MSRE* heat exchanger system, focusing on design considerations of the MSRE heat exchanger, considering that vibration could be a problem and that flow tests should be conducted with the heat exchanger (Kedl and McGlothlan, 1968).

The data presented in the following tables are derived from these documents. Table 1 summarises the key design and operational parameters of the MSRE, while Table 2 and Table 3 outline the thermo-physical properties of the MSRE fuel salt and graphite moderator, respectively, used in this study. To achieve criticality in zero-power physics experiments on the MSRE, fuel salt composition was set to 62.5% LiF, 31.6% BeF2, 5.1% ZrF4, and 0.8% UF4 (by molar percentages), and a mass fraction of ^{235}U in the salt to 1.408 $\pm$ 0.007 wt% (Forsberg, 2007).

2.2. Numerical approach and code overview

2.2.1. GeN-Foam

The GeN-Foam (Generalised Nuclear Foam) (Fiorina et al., 2015) open-source software is a solver for the steady-state and transient analysis of nuclear reactors, it can be used for pin-type, plate-type, or liquid fuel of reactors featuring pin-type, plate-type, or liquid fuel. With the option to select the physics to solve at runtime, it includes sub-solvers for neutronics, single- and two-phase thermal hydraulics, and thermal mechanics. For neutronics, thermal-hydraulics, and thermal-mechanics, three distinct meshes are used.

GeN-Foam performs coupled multi-physics simulations through an iterative exchange between its neutronics and thermal-hydraulics sub-solvers. The neutronics sub-solver operates on a dedicated computational domain to predict the volumetric power density based on parameters such as fluid temperature and density, structural temperatures, and, when applicable. This fuel displacement, obtained from the thermal-mechanics sub-solver, is used to account for expansion-related feedbacks in the reactor core. These fields are linked to reactivity feedback via pre-tabulated, temperature-dependent nuclear cross-sections. The resulting power density is then passed to the thermal-hydraulics sub-solver, which solves for key quantities including fluid temperature, density, velocity, and structural temperatures. In the case of liquid-fueled reactors like MSRs, the velocity field is also used to model precursor advection. The thermal-hydraulics sub-solver supports both single- and two-phase flows and can handle porous-media regions (e.g., core graphite) using empirical models for heat transfer and pressure loss, as well as fine-mesh CFD-like domains for regions such as plena and downcomers. This modular structure allows accurate simulation of power generation and heat transport in complex geometries (Fiorina, 2022).

Fig. 2 shows the GeN-Foam workflow, illustrating the integration of thermal-hydraulics and neutron diffusion for analysing graphite temperature distribution in the MSRE.

The GeN-Foam multi-physics solver has already been used for liquid-fuelled reactor systems such as (Bao, 2016; He et al., 2019; SHI, J., 2020), but in this work, a full 3D analysis for analysing the MRSE coupled multi-physics is done for the first time.

This approach captures the immediate effects of neutron flux and gamma heating on the graphite. However, it does not dynamically account for long-term material degradation caused by cumulative neutron fluence, such as reduced thermal conductivity and changes in structural properties. While the initial energy deposition reflects fluence effects, time-dependent degradation of graphite properties from sustained exposure is not included in the model.

2.2.2. Neutronics modelling

In contrast to solid fuel reactors, in liquid fuel reactors, it is necessary to consider the movement of delayed neutron precursors (DNP) with the flowing fuel, requiring additional terms in the governing equation to represent their movement and diffusion (Krepel et al., 2005). DNPs generated in the core can decay elsewhere in the primary loop, impacting the local neutron balance and reactivity. This leads to a strong non-linear coupling between fuel motion and neutron dynamics (Cammi et al., 2011). The impact of delayed-neutron precursor migration in MSRs is demonstrated by e.g. (Davidson et al., 2024; SHI, J., 2020; Yamamoto et al., 2006).

In this study, the GeN-Foam neutron diffusion solver is used, as diffusion theory is expected to work well for approximating neutron behaviour due to the relatively uniform neutron flux of MSR cores.

The cross-sections used are derived from a previous work by Bao (Bao, 2016) in which Serpent-2 and the JEFF-22 data library were used to generate the zone-averaged cross-section sets needed by the GeN-Foam neutronic solver. A vertical cutaway view of the model in Fig. 2 shows the core, lower plenum, upper plenum, downcomer, and control rods modelled in Serpent-2. During critical operation, with the fuel salt

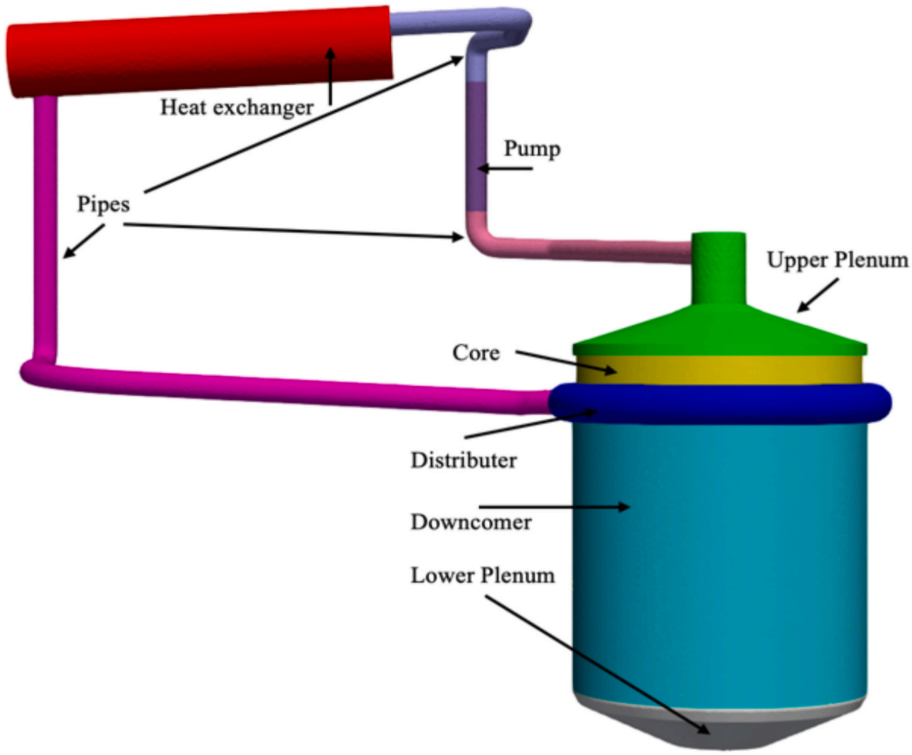

Fig. 3. Detailed view of MSRE model primary loop.

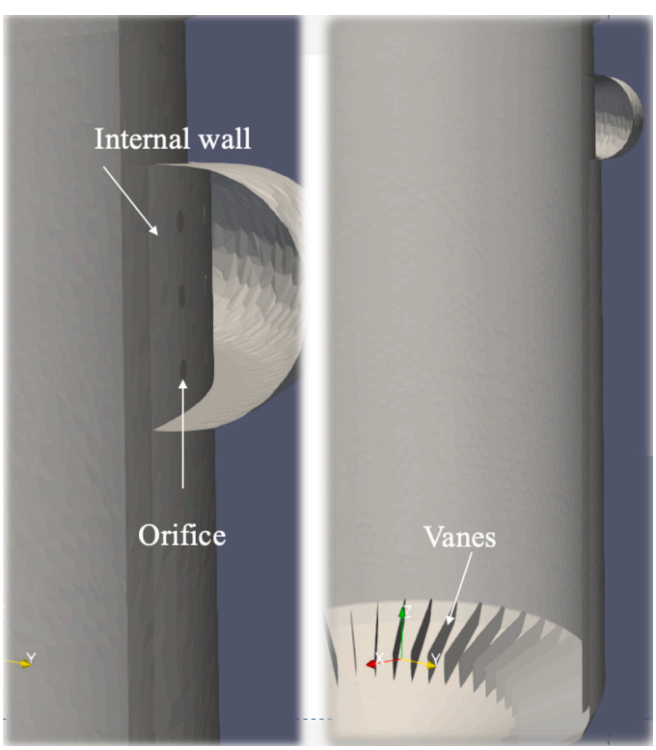

Fig. 4. Detailed views of key MSRE components. (Left) Orifices within the flow distributor. (Right) Vanes located in the lower plenum.

stationary, the reactor configuration involves two rods fully withdrawn at 129.54 cm from the top position of the core, while the third rod is inserted to 3% of its integral worth, at 118.36 cm (Shen and Fratoni, 2020).

Explicitly defined material properties and material-specific nuclear data libraries are used to account for temperature and density effects. Whereas structural and shielding materials use 300 K libraries for temperature, regions like the fluid fuel in the core, upper plenum, and lower plenum use libraries equivalent to 900 K. Thermal scattering treatment at 900 K is a feature of the graphite moderator. Mass densities, such as 2.252 g/cm^3 for fluid fuel, are specified explicitly in the input.

Regarding the feedback from thermal analysis, in this study, a simplified parametric approach was used. Temperature-dependent nuclear cross-sections were generated using Serpent at two representative temperatures: 900 K and 1200 K. The base case, which reflects the actual reactor conditions, used the 900 K nuclear data. A perturbed case was simulated using cross-sections corresponding to 1200 K. Although cross-section updates were not performed continuously during the simulation, this approach captures the main feedback effects and provides a basis for future improvements, such as incorporating cross-section updates.

2.2.3. Porous-media modelling

To reduce computational time of the simulation in this study, the core, heat exchanger, and inner distributor are modelled as porous zones. Void and volume fractions for the porous zones are based on the ORNL-TM-730 report data. The core was modelled as a single cylinder, rather than discretely modelling the fuel channels. A coarse mesh and porous medium approach was used, and control rods were not included in the model. The core's volume fraction is approximately 0.775 for the graphite structure, with the remaining 0.225 filled by fluid fuel (Batara, 2020; Vafai, 2015). The graphite stringers are 170.03 cm long and are arranged in a vertical, close-packed array. There are a total of 1140 full-size equivalent passages, including fractional sizes (Haubenreich et al., 1964).

The porous medium approach was also used for the calculation of the heat exchanger in which only the shell is modelled. The void and volume fractions were based on dimensional data from the ORNL-TM-2098 report (Fratoni et al., 2020). It was determined that approximately 40% of the volume was occupied by the tube structure, while the remaining 59.4% was filled with fluid fuel.

For fine mesh calculations, orifices are included in the distributor to

Table 4
Geometric and Modelling Parameters.

Component	Dimensions (cm)
Core	Inner diameter: 70.49
Reactor vessel	Inner radius: 73.66; thickness: 1.43–2.54
Distributor	Torus cut; inner radius: 10.16; thickness: 0.82
Outlet Pipe	Cylinder; radius: 12.70; thickness: 2.49; length: 35.87
Heat Exchanger	Inclined cylinder; diameter: 40.64; length: 199.27; 3° downward tilt
Pump	Vertical pipe; diameter: 12.70; length: 80.0

closely match the MSRE. In coarse mesh calculations, to reduce computation time, the distributor is divided into two zones, and the orifices are not modelled. Instead, the inner zone is given higher porosity to simulate the pressure drop. The volume fraction was adjusted to match the desired pressure drop, consistent with what a detailed orifice structure would yield.

2.3. Geometry and mesh

2.3.1. Geometry

The MSRE core geometry is reproduced using the SALOME platform's geometry module (Salome platform, 2025) to model the reactor's primary loop, including the core, upper plenum, lower plenum, distributor volute, downcomer, pipes, heat exchanger, and pump. Extensive documentation on the MSRE is available, with key data sourced from public domain publications (Fratoni et al., 2020; Prince et al., 1968; Robertson, 1965; Shen et al., 2021). The reactor geometry was mainly constructed using dimensional data from ORNL-TM-728 (Robertson, 1965), supplemented by Bao's work (Bao, 2016).

Fig. 3 shows a detailed view of the primary loop, while the following subsections outline the reactor core data used in the geometry construction.

The 48 anti-swirl vanes in the lower plenum prevent the tangential motion of the fuel, directing it into the plenum centre for vortex-free flow. In the geometry model, these vanes are represented as zero-thickness baffles. Fig. 4 shows the SALOME-generated geometry model, highlighting the vanes in the lower plenum. An external torus cut flow distributor is placed atop the downcomer that distributes the fluid fuel and passes through the orifices that are used to obtain a uniform angular flow distribution to the downcomer (Kedl, 1970).

As shown in Table 4, a summary of the geometric dimensions and modelling parameters for the MSRE components is provided.

2.3.2. Mesh

The ANSYS-Fluent Mesh generator was chosen for this study for its Mosaic meshing method. Three meshes were created to assess spatial convergence: a fine mesh with 1400×10^3 cells, a medium mesh with 600×10^3 cells, and a coarse mesh with 290×10^3 cells for simulating the MSRE's primary loop. Some parameters, such as velocity and pressure at specific locations (particularly where comparison with ORNL data was possible) were evaluated across all three meshes. Since the results of the three approaches were closely aligned, a coarse mesh approach was used for the simulation.

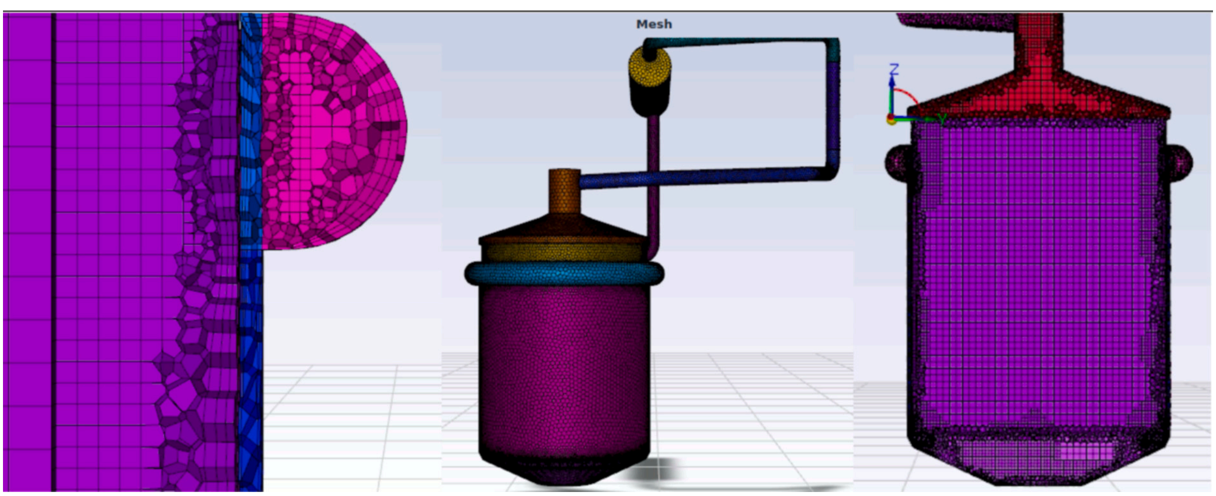

Fig. 5. Volume mesh structure using the poly-hexcore method.

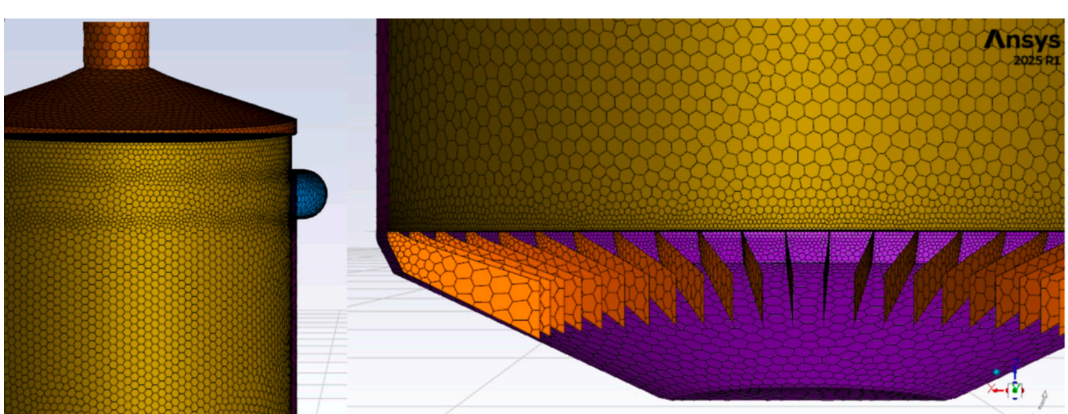

Fig. 6. Surface mesh configuration of the MSRE primary loop.

Table 5
Coarse mesh specifications.

Parameter	Value	Parameter	Value
Surface mesh min size (m)	0.015	Boundary layer growth rate	1.2
Surface mesh max size (m)	0.045	Volume mesh Buffer layers	3, 1
Surface growth rate	1.2	Volume mesh peel layers	2, 1
Curvature normal angle (deg)	18	Volume mesh max	0.04
Max gap distance for join-intersect (m)	0.009	Volume mesh Min	0.02
Number of boundary layers	3	Number of cells	290,192
Transition ratio on boundary layer	0.272	Min quality cell	0.1000508

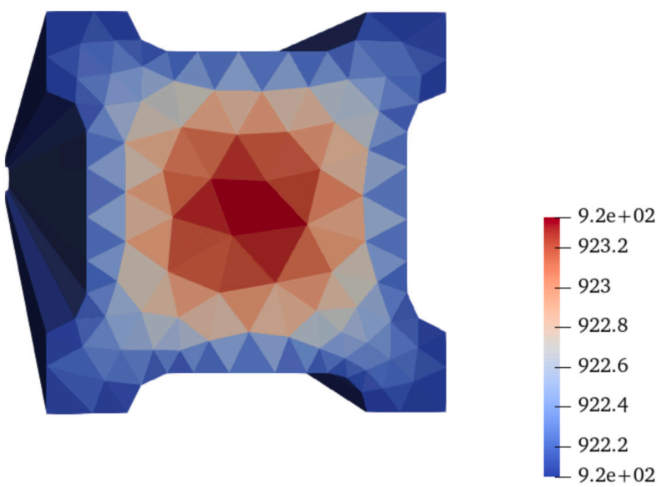

Fig. 7. Temperature distribution in single graphite slab.

In this work, the Poly-Hexcore method was employed to generate the volume mesh. This approach reduces the total element count by about 20–50% compared to traditional hexcore meshes. In this method, polyhedral elements were used in the transition zone, while hexahedral elements were employed in the core (Krishna et al., 2019). The surface mesh configuration used for the domain boundaries is shown in Fig. 6, while the volume mesh structure, illustrating the internal polyhedral and hexahedral arrangement, is presented in Fig. 5. Key mesh generation parameters, such as element sizing, growth rates, boundary layer settings, and quality metrics, are summarised in Table 5.

2.4. Physical models and boundary conditions

2.4.1. Boundary conditions

The simulations are performed using a 3D model of the full MSRE primary loop. The system is initialised with a uniform temperature of 908 K and an internal pressure of 0.134 MPa. A fixed flux pressure boundary condition is used to maintain this pressure during the simulation. To match the thermal behaviour, the temperature is increased gradually over time to reach the point that fuel temperature feedback is affected, after lowering the value it holds steady. The reactor vessel boundaries are treated as walls, with zero-gradient boundary conditions applied for temperature on all walls and internal baffles. The vessel walls are considered adiabatic. Turbulence effects near walls are handled using wall functions for turbulent kinetic energy, dissipation rate, and eddy viscosity. Fluid flow across the loop is driven by a momentum source placed in the vertical pipe representing the pump. This is tuned to achieve the design flow rate. For the heat exchanger, a simplified model is used, a straight cylinder tilted 3 degrees downward from the inlet to the outlet. A constant temperature boundary condition is applied here to simulate cooling. Since the graphite structure isn't modelled explicitly, the core region is treated as a porous zone. The porous media properties (like internal heat generation and thermal conduction) are based on the actual graphite geometry and power profile.

In the neutronics part, fixed-value boundary conditions are used for neutron flux at all component boundaries. For delayed neutron precursors, zero-gradient boundary conditions are applied on the baffles and walls.

2.4.2. Power modelling

To capture the intricate thermal behaviour of graphite in the MSRE, a lumped-parameter power model was chosen. This model's nodal flexibility allows for detailed temperature distribution representation while incorporating of power density and heat conductance factors into the energy balance equation. Solving for n nodes requires n + 1 conductances H and yields n + 1 temperatures T_i, which act as state variables. T_{max} and $T_{surface}$ are subsequently calculated. The governing equation for each node's behaviour is based on a fundamental energy balance, considering the i^{th} node:

$$V_i \rho_i Cp_i \frac{dT}{dt} = VQqf + H[i](T_{i-1} - T_i) - H[i+1](T_i - T_{i+1})[1] \quad (3)$$

where V_i represents the volume of the ith node, ρ_i is the density, and Cp_i is the specific heat capacity of the material at node i. Q represents the power density from neutronics, assumed to distribute uniformly across the structure, and the power fraction qf concentrates this power over specific nodes.

To derive the heat conductance values, temperature differences between the maximum, average, and surface temperatures of a single graphite stringer were modelled. This involved creating a CAD model and mesh of a single stringer using the SALOME mesh generator and a custom solver in OpenFOAM was used to compute the maximum and surface temperatures based on convection, applying the energy conservation equation.

After convergence, the average and maximum temperatures of the graphite were found, with a boundary temperature of 922 K. Using these temperatures and the derived equation, the heat conductances for the MSRE graphite were calculated for both the maximum temperature and surface temperature (see Fig. 7).

In line with ORNL-TM-0378 (Engel and Haubenreich, 1962), a 0% fuel permeation in the graphite was assumed for this simulation under a 10-MW reference condition. This calculation assumes that 6% of the reactor power is generated in the graphite in the absence of fuel permeation, based on gamma and neutron heating estimates by C.W. Nestor (Engel and Haubenreich, 1962).

2.4.3. Physical models overview

The selection of physical models in this study was guided primarily by the ORNL report TM-3229, along with other relevant references on molten salt reactor behaviour. Table 6 provides an overview of the models used to represent turbulence, drag, and heat transfer in the MSRE system. These models were chosen to reflect the specific flow and thermal characteristics of molten salt under MSRE conditions and were implemented within the Gen-Foam framework used for the simulations.

3. Results and discussion

3.1. Neutronic analysis

The Serpent-based model of this study, described in Section 2.2.2 yields a k_{eff} value of 1.02186 ± 0.00129. Comparison with previous studies shows notable variations in the predicted multiplication factors. For example, Bao reported a Serpent-2 value of 1.15799. This higher value can be attributed to several modelling differences: Bao did not

Table 6
Summary of physical models used in the simulation.

Model Type	Approach/Model	Key Equation	Reference
Turbulence Model	Standard porous $k-\varepsilon$ model and RANS with Boussinesq's hypothesis	–	(Kedl, 1970)
Flow Assumption	Incompressible flow	–	(Kedl, 1970)
Drag Model	Generalised drag model using Blasius correlation	$f_d = 0.316 \times Re^{0.25}$ (6)	(van der Linden, 2012)
Heat Transfer Model	Nusselt-Reynolds-Prandtl-Power model (Gen-Foam implementation)	$Nu = 0.023Re^{0.8}Pr^{0.4}$ (7)	(White, 2011)
Heat Transfer Type	Convective and conductive (radiation excluded in fluid flow but included in total energy transfer)	–	(Kedl, 1970)

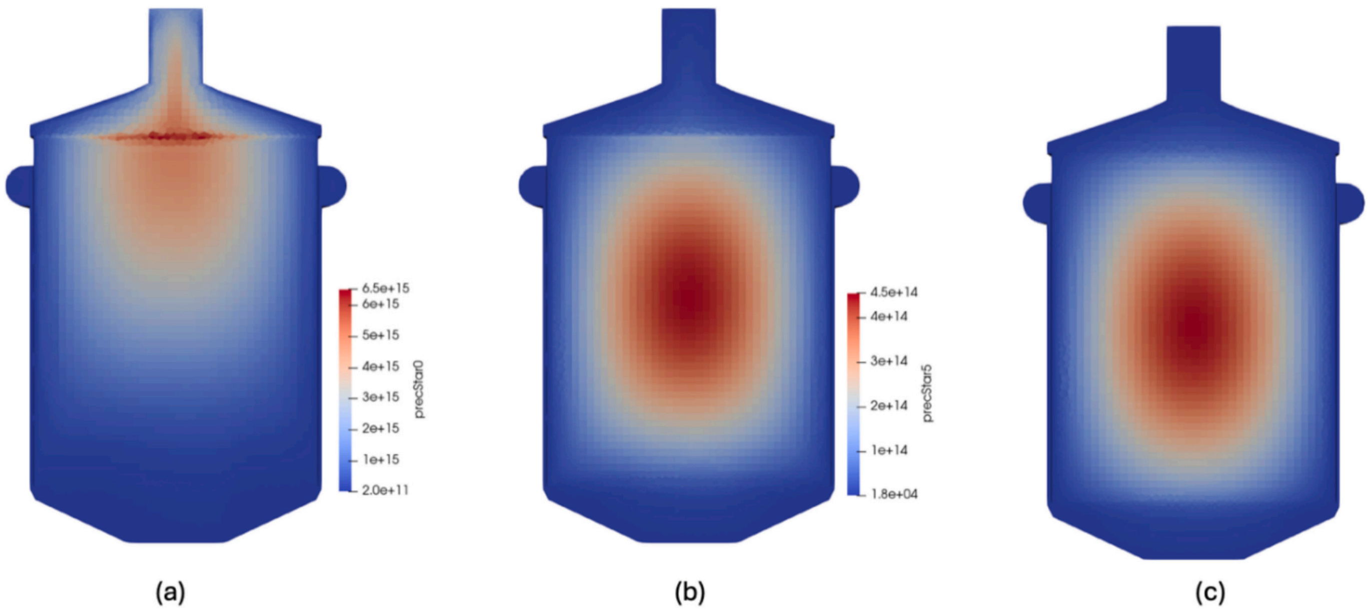

Fig. 8. Distribution of precursor concentrations: (a) with the longest half-life, (b) with the shortest half-life, and (c) without fuel movement.

include the central graphite block, incorporated control rods differently, and employed the JEFF-3.2.1 nuclear data library, all of which affect reactivity predictions (Bao, 2016).

Shen et al. obtained a Serpent-calculated k_{eff} of 1.02132. Their study examined several geometric configurations, and even the case most similar to the present model involved key differences: the central block basket was not homogenised, and both the upper and lower plena were modelled as hemispheres. Additionally, their work relied on the ENDF/B-VII.1 cross-section library rather than the JEFF-22 dataset used in this study (Shen et al., 2021).

ORNL-TM-935 reports a k_{eff} value of 0.91 (Engel et al., 1964). This significantly lower value reflects the fact that the calculation was performed under subcritical steady-state conditions. The purpose of that analysis was not to evaluate operational reactivity but to determine the necessary external neutron source strength and detector sensitivity for reactivity monitoring. Consequently, its results are not directly comparable to full-power criticality calculations.

In the neutronic analysis of this study, GeN-Foam produced an effective multiplication factor (k_{eff}) of 1.08164. It is important to highlight that control rods were not explicitly modelled in GeN-Foam. The observed discrepancy between GeN-Foam and Serpent results of this study can be attributed to the absence of explicit control rod representation in the GeN-Foam simulations. Bao's corresponding GeN-Foam calculation produced a k_{eff} value of 1.11446 (Bao, 2016).

To assess the impact of control rods on reactivity, a benchmark control rod bank worth of 5611 pcm as reported by (Haubenreich et al., 1964), was added to the Serpent results of the current study.

When the control rods are inserted, the reactivity ρ was calculated as follows:

$$\rho_{with\ control\ rod} = \frac{k_{eff} - 1}{k_{eff}} = \frac{1.021860 - 1}{1.021860} \approx 0.021390 \tag{8}$$

By adding the control rod worth to the reactivity with the control rods inserted, the resulting reactivity in the absence of control rods was determined:

$$\rho_{without\ \ control\ rod} = \rho_{with\ control\ rod} + 0.05611 = 0.07750 \tag{9}$$

This indicates a notable increase in reactivity in the absence of control rods, leading to a new k_{eff} calculated as follows:

$$k_{eff,without\ control\ rod} = \frac{1}{1 - \rho_{without\ \ control\ rod}} = \frac{1}{1 - 0.07750} \approx 1.08394 \tag{10}$$

The calculated values indicate a significant increase in k_{eff} from 1.021860 to 1.08394 when the control rods are removed. This value more closely aligns with the current study Gen-Foam results of 1.08164.

While the k_{eff} value of 1.02186 aligns with the expected reactor performance with control rods inserted, the differences in k_{eff} calculations between the two frameworks can be explained by their respective modelling assumptions. In the GeN-Foam model, control rods were not included, which means the control rod worth was not reflected in the computed k_{eff}, leading to a higher reactivity estimate. In contrast, the Serpent model accounted for the presence and positioning of control rods, which reduces k_{eff} accordingly, although it did not model fuel movement. If the control rod worth were included in the GeN-Foam simulation, the resulting k_{eff} would align more closely with the values obtained from the Serpent model.

The steady-state precursor distribution in the reactor vessel is shown in Fig. 8. Specifically, Fig. 8 (a) and (b) illustrate r-z distributions at the vessel mid-plane of the two precursor groups with the longest and shortest half-lives, respectively, while panel (c) shows the precursor

Table 7
Velocity values from ORNL report and corresponding simulation results.

Parameter	Simulation Value	MSRE Data (ORNL)	Deviation (%)
fuel velocity in core (m/s)	0.2115	0.2133	0.84
fuel velocity in Upper head (m/s)	0.1580	0.1524	1.46
fuel velocity in inlet pipe (m/s)	6.08	5.85	3.93
fuel velocity immediately entering distributor (m/s)	6.88	7.01	1.85
fuel velocity 76 cm up in downcomer (m/s)	2.09	2.19	4.52

distribution without fuel movement. A noticeable discontinuity in the precursor distribution can be observed between the core region and the upper head area. This discontinuity is due to the difference in porosity between these two regions; if the porosity were uniform throughout, such a discontinuity would not be present.

To provide support for the results, a code-to-code comparison is made with a previous 2D simulation of the MSRE using 23 energy groups, which analysed delayed neutron precursor (DNP) distributions for groups 1, 3, and 6 under both static and circulating fuel conditions (SHI, J., 2020). In the static case, the DNP distribution closely follows the total neutron flux since the precursors do not move, and their concentration remains proportional to the local neutron production. However, under circulation, significant redistribution occurs, which has the smallest decay constant and a half-life greater than 50 s. This allows it to spread across the core and external loop before decaying. As the decay constant increases (and the half-life decreases), the spatial spread of precursors becomes more limited. The trends observed in our current results are consistent with this reference study.

3.2. Thermal-hydraulic analysis

The objective of the thermal-hydraulic analysis of this study is to predict the steady-state pressure and velocity distributions within the reactor components. The following results are obtained according to the procedure described in Section 2.4.

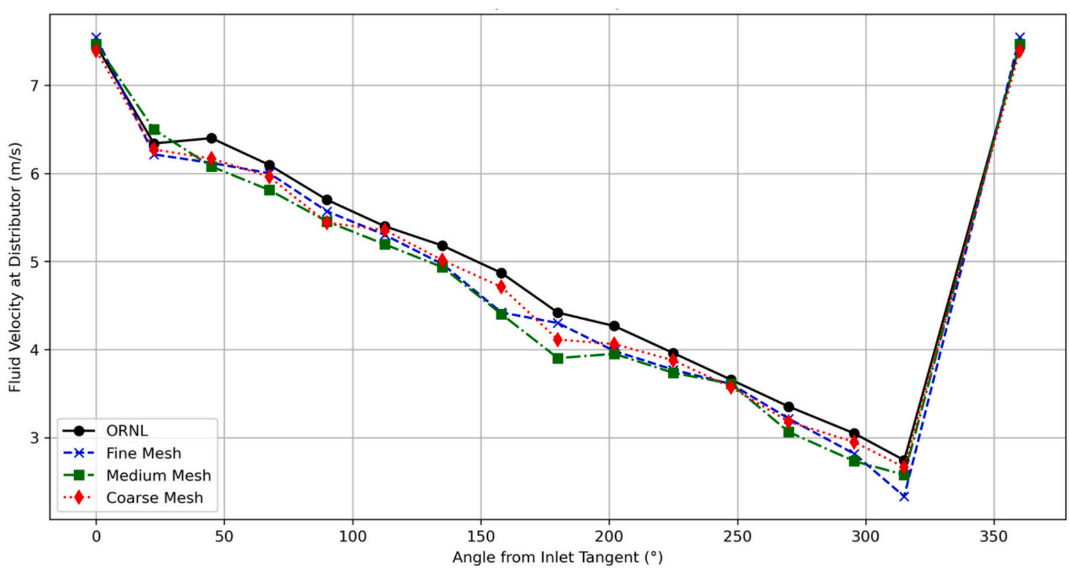

Fig. 9. Angular distribution of fluid velocity at center line of the distributor.

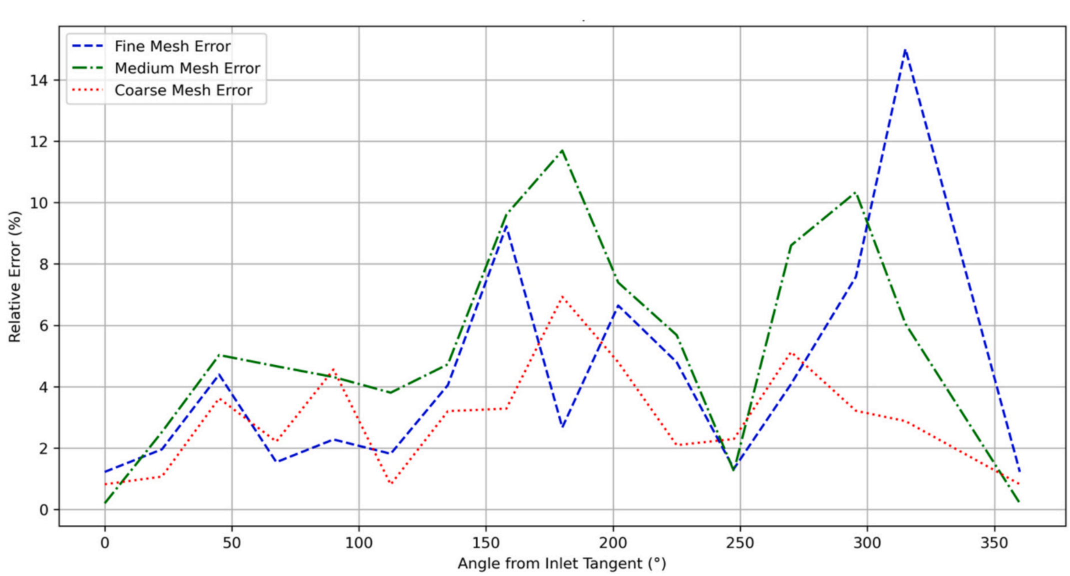

Fig. 10. Relative error in simulated axial velocity profiles compared to ORNL data.

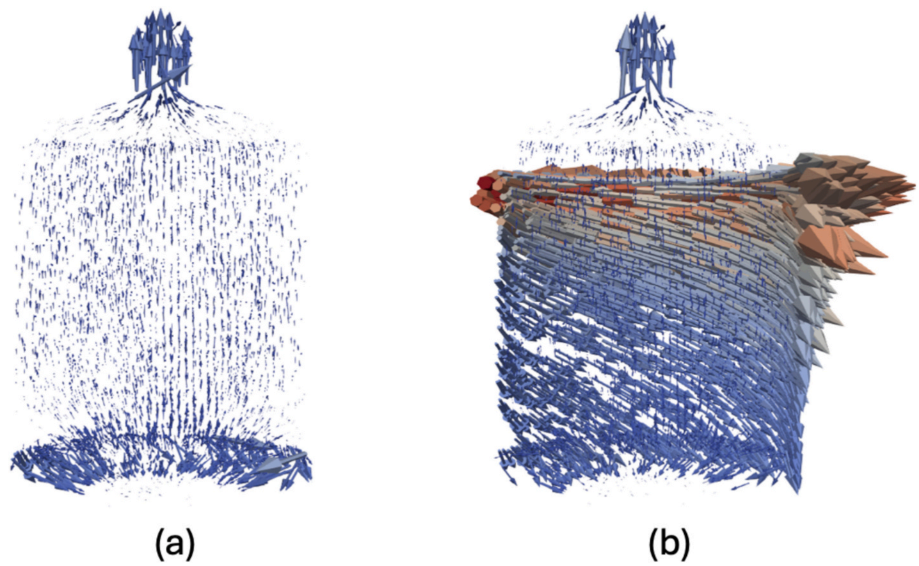

Fig. 11. Distribution of the fluid fuel inside the vessel. **(a)** Velocity distribution in the lower plenum, core, and upper plenum, highlighting core flow patterns and swirl suppression by the anti-swirl vanes. **(b)** Velocity distribution across the entire reactor vessel, showing flow rotation through the distributor and velocity reduction in the distribution chamber.

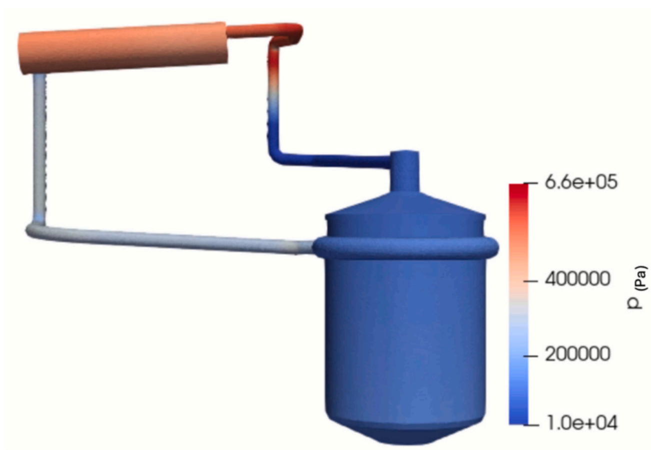

Fig. 12. MSRE pressure distribution visualisation.

3.2.1. *Flow field*

Detailed fluid-dynamic data of MSRE are documented in ORNL-TM-3229 (Shen and Fratoni, 2020). These data were used as a reference to validate the results obtained with the GeN-Foam model in this study. Table 7 presents a summary of the comparison for the fuel velocity values at different locations, like the salt velocity through the moderator fuel channels inside the core, 5.08 cm distance from the wall in the upper head, and fuel velocity 76 cm up in the downcomer. Taking into account the modelling simplifications made in this study by adopting a porous-medium method to represent sections like the core, heat exchanger, and pump. For simplicity, the distributor's inner wall was replaced by orifices with a porous region, resulting in two porous zones with specified porosity. This is the reason for the most notable deviation, which occurs for velocity 76 cm up the downcomer. The primary goal of Table 7 is to validate that the boundary conditions were applied appropriately and that the model replicates a physically consistent flow distribution, not just velocity.

Fig. 9 shows the velocity distribution within the distributor, comparing the results from the ORNL-TM-3229 (Kedl, 1970) report with those from Gen-Foam, Simulation results for different mesh resolutions are compared with ORNL benchmark data to assess mesh sensitivity.

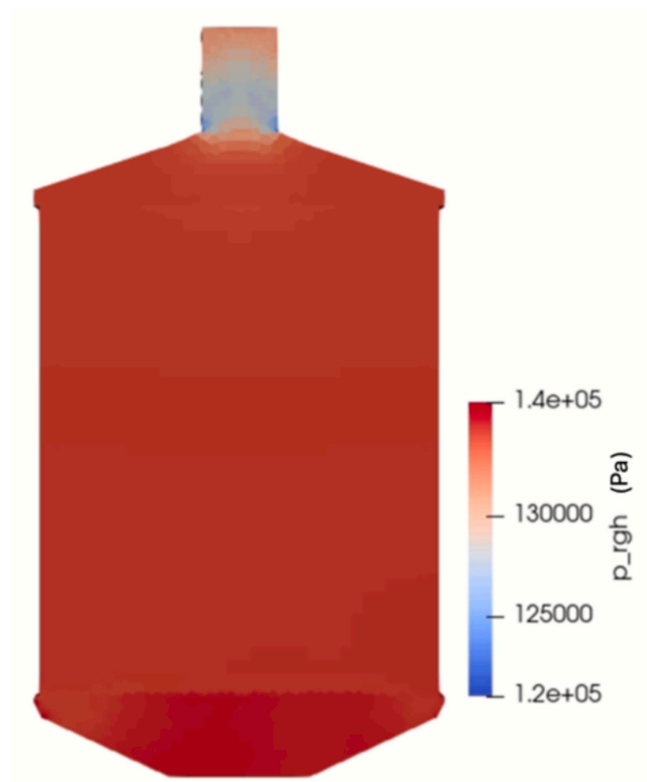

Fig. 13. Core plenum pressure profile.

Table 8
Quantitative comparison of simulated pressure values and MSRE.

Parameter	Simulated Value (Pa)	MSRE Recorded Value (Pa)	Deviation (%)
Pressure at MSRE core	133,176	134,473	0.964
Pressure at heat exchanger outlet	483,842	479,212	0.966

Table 9
Temperature Comparison: Fuel & Graphite Mean, Simulation vs. MSRE Data with Deviation.

Temperature (K)	Simulation Value	MSRE Data (ORNL)	Deviation (%)
Fuel inlet	908.51	908.15	0.039
Fuel outlet	936.003	935.921	0.008
Graphite mean	935.61	936.42	0.08

Fig. 10 presents the relative error in the simulated axial velocity profiles compared to the ORNL-TM-3229 measurements, highlighting how the mesh resolution influences accuracy within the distributor.

The coarse mesh uses a simplified representation of the distributor in which the inner region is modelled as a porous zone in Section 2.2.3. This approach smooths the flow and removes the jetting and small recirculation features created by the orifices. As a result, the velocity field from the coarse mesh appears smoother and can produce a lower point-wise error, even though the physical detail is reduced.

In contrast, both the medium and fine meshes explicitly resolve the inner wall and the orifices. The fine mesh captures sharper velocity gradients, small separation bubbles, and detailed jets around the holes that the medium mesh can only partially resolve. These additional small-scale features affect the local velocity distribution and can increase the point-wise relative error when compared with the smoother experimental profile. Thus, the higher error observed in the fine mesh does not indicate poorer physical accuracy; rather, it reflects the fact that the fine mesh resolves more of the true flow structures, while the medium and coarse meshes smooth them out.

To give a clearer view of the fluid fuel velocity distribution, the two images highlight different aspects of the flow. Fig. 11(a) focuses on the fluid fuel velocity distribution in the lower plenum, core, and upper plenum, offering a closer look at the uniform velocity patterns and the rotational flow around the vanes inside the core. By leaving out the surrounding structures, this view emphasises the core flow dynamics. The fluid fuel initially exhibits rotational motion as it circulates through the downcomer region. As the fluid reaches the anti-swirl vanes, this rotation is actively suppressed, as the vanes are designed to reduce tangential flow and guide the fluid more uniformly toward the core inlet. Additional rotation occurs around these vanes, which are specifically designed to suppress swirling motion and promote a more uniform inlet flow into the core, and, ultimately, the homogeneous movement of the fluid fuel through the core. On the other hand, Fig. 11(b) shows the fluid fuel velocity distribution across the entire reactor vessel, including

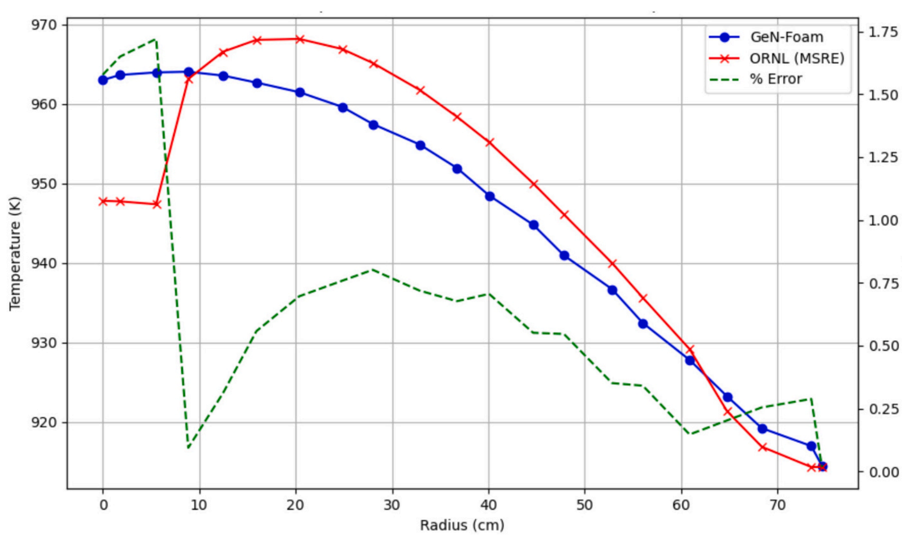

Fig. 14. Radial temperature profile in MSRE core near midplane.

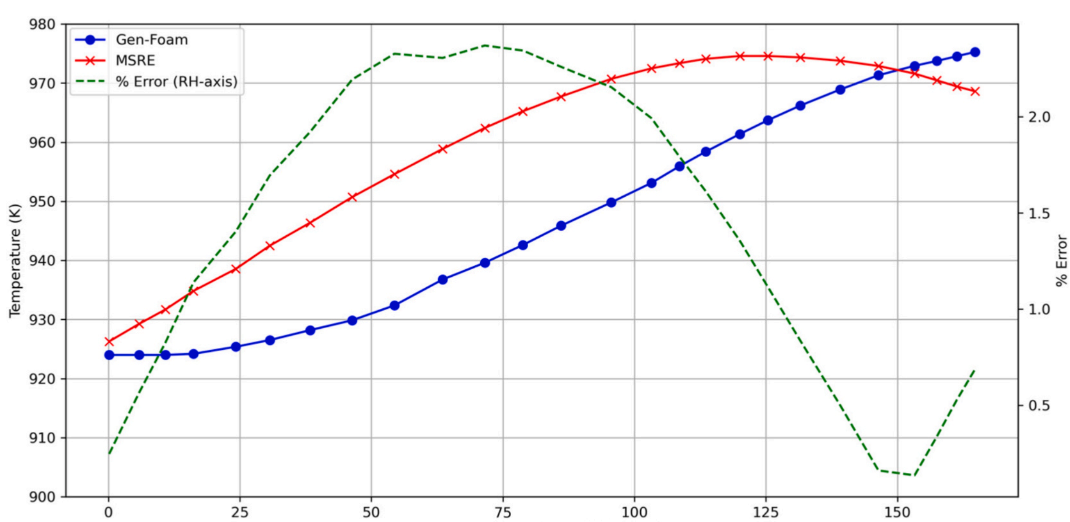

Fig. 15. Axial temperature profile in MSRE core.

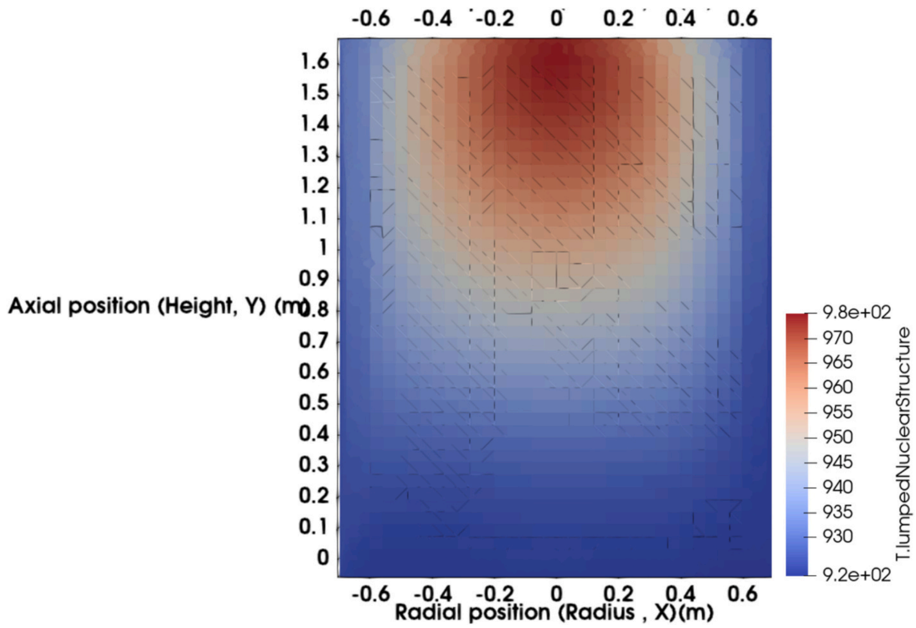

Fig. 16. Lumped nuclear structure temperature distribution in MSRE core.

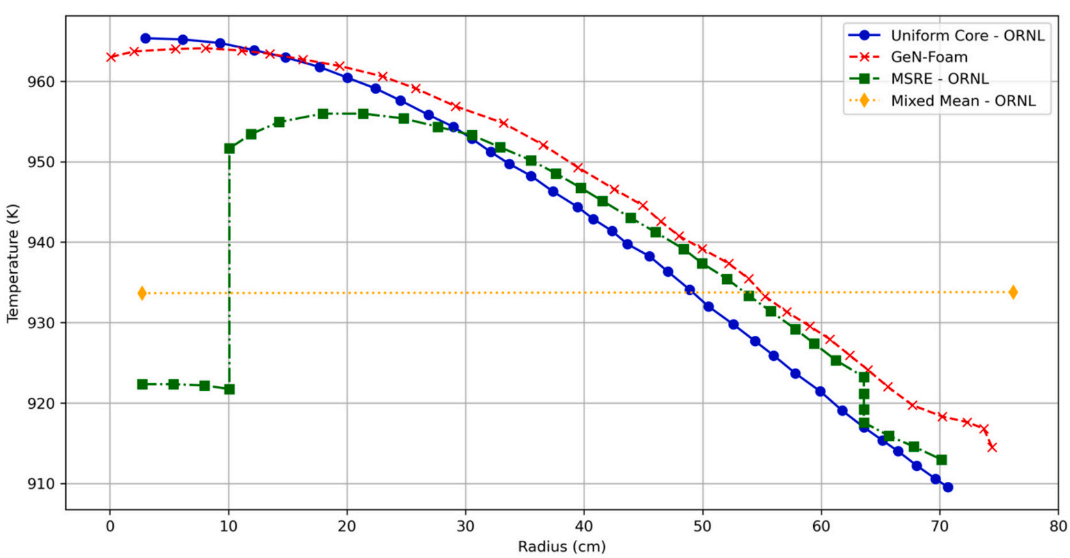

Fig. 17. Channel outlet temperature for MSRE and for uniform core.

the volute and other peripheral components, giving a complete picture of how the flow behaves throughout the system. The figure illustrates the smooth rotation of the flow through the distributor and the movement of the fluid with velocity loss in the distribution chamber.

3.2.2. Pressure field

The pressure field distribution under steady-state conditions was analysed, with results shown in Fig. 12 and Fig. 13. The primary pressure was maintained consistent with the MSRE's standard operating conditions. The spatial pressure demonstrates variation and shows a smooth gradient, indicating stable flow dynamics throughout the porous medium. Table 8 compares simulated pressure values with MSRE recorded data at two key locations: the core reference point and the heat exchanger outlet.

3.2.3. Steady-state graphite temperature during

To model the graphite temperature distribution within the MSRE under steady-state conditions, it was essential to consider the coupled effects of neutronics and fluid dynamics. The analytical solution for graphite temperature analysis is described broadly in ORNL report (Kedl, 1970). In this study, the temperature distribution of the graphite was determined by simultaneously solving the neutronics and thermal-hydraulics equations. The graphite temperature changes over time, eventually stabilising when equilibrium is reached.

Table 9 compares temperature values from ORNL studies with results from GeN-Foam simulations in this study.

Fig. 15 illustrates the lumped temperature distribution within the MSRE core, showing a gradual increase in temperature influenced by the structural effects on the graphite.

Fig. 14, Fig. 15 and Fig. 16 show the radial and axial temperature distributions within the graphite at the midplane and the hottest radial position. The radial distribution is presented in Fig. 14 and the axial distribution in Fig. 15. The comparison between the ORNL data and the Gen-Foam shows results with a maximum error of 2.5%. At the reactor's

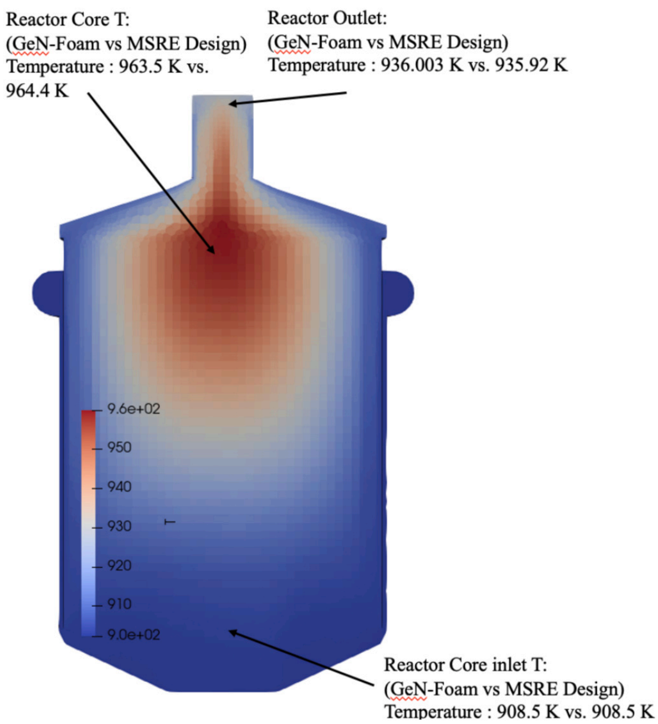

Fig. 18. Overall temperature distribution within the MSRE core.

Table 10
Comparison of simulation and ORNL reported temperatures.

Temperature Point	GeN-Foam (K)	ORNL Design (K)	Deviation (%)
Core Inlet Temperature	908.5	908.5	0.000%
Core Outlet Temperature	936.003	935.92	0.009%
Top of Graphite Matrix (~167 cm)	963.5	964.4	0.093%

design power level of 10 MW, with inlet and outlet temperatures of 908.6 K and 936.0 K, respectively, the simulations showed the average temperature of the graphite in the reactor vessel, excluding the volute, was calculated to be 935.6 K, close to the ORNL reported value of 936.4 K.

The radial temperature distribution within the MSRE core, presented in Fig. 14, shows a MAPE of 0.62% when compared with the ORNL data, and the axial temperature distribution, shown in Fig. 15, yields a MAPE of 1.09%.

The temperature distribution of an idealised uniform core, derived from ORNL-TM-0378, is shown in Fig. 17. This core features a consistent fuel fraction and uniform fuel velocity, with a radial power distribution following a Bessel function that vanishes at the inner core-shell. It is normalised to the same inlet and mixed-mean outlet temperatures as those calculated for the primary MSRE core. This idealised uniform core was used as a reference because its description aligns with the porous media approach employed in the modelling of the MSRE reactor. The homogeneous temperature distribution, shown in Fig. 18 and compared with the data from ORNL-TM-0378 (Engel and Haubenreich, 1962).

Fig. 18 provides an overview of the temperature distribution within the MSR core, including the upper head region. To support validation, selected temperature points from the GeN-Foam simulation are compared with ORNL design values in Table 10. The comparison shows values at the core inlet and a small deviation near the top of the graphite matrix, while the outlet temperature is slightly higher in the simulation than in the reference data.

4. Conclusions

In MSR reactors circulation of liquid fuel leads to strongly coupled physical phenomena. The circulation of delayed neutron precursors modifies the neutron flux distribution by impacting the spatial profile of volumetric heat generation. This thermal feedback alters the local temperature field, leading to temperature-dependent variations and consequently influencing the thermal-hydraulic behaviour of the molten salt.

The strong coupling between neutronics, thermal-hydraulics, and fuel transport phenomena necessitates the use of a multi-physics modelling approach to represent reactor behaviour. The Gen-Foam framework provides this capability by incorporating the coupled effects of heat transfer, fluid flow, and neutron transport, enabling high-fidelity simulation of molten salt reactor (MSR) dynamics.

In this study, we simplified the modelling of complex zones, including the pump and heat exchanger, by using a porous media approach. This method also allows for the replacement of certain internal parts, such as the graphite structures, with porous zones, allowing the assignment of porosity and relevant physical properties. That made it possible to use a coarser mesh in those regions, thereby **reducing computational time in** the simulation without losing important thermal-hydraulic effects.

This study implemented and verified a multi-physics model of the MSRE in Gen-Foam, primarily focusing on predicting the graphite temperature distribution.

Verification tests were primarily conducted to compare the current model against the fluid dynamics data of the MSRE reactor as presented in the ORNL-TM-3229 report. With core pressure and fuel velocity results exhibited deviations of 0.96% and 0.84%, respectively.

Additionally, coupled neutronics and thermal-hydraulics simulations were performed to compare the model's predictions with the temperature calculations for the MSRE core during steady-state power operation, as reported in ORNL-TM-0378. These simulations yielded a Mean Absolute Percentage Error (MAPE) of 0.62% for radial and 1.09% for axial graphite temperature distribution, compared to ORNL-TM-0378 data, The overall fuel temperature MAPE was 0.51%, demonstrating the model's accuracy in capturing thermal-hydraulic behaviour.

The homogenised porous media approach, combined with advanced multi-physics modelling techniques, produced results in good agreement with the expected temperature distribution and thermal characteristics.

This study provided a 3D model of the MSRE, offering improvements over previous models. Meshing complexities and high computational costs were identified as key challenges, suggesting that 2D modelling approaches could offer a more computationally efficient alternative for initial analyses.

Additionally, discrepancies in neutron flux distribution between Gen-Foam and ORNL/Serpent data indicate the need for more detailed modelling techniques to improve flux predictions.

In conclusion, a 3D model was developed and verified using ORNL benchmark data, demonstrating good agreement with the reference results. This supports the applicability of the model for simulating molten salt reactor (MSR) behaviour. Additionally, the input data from this 3D model can be utilised to construct a corresponding 2D model, enabling comparative analysis. These developments support the broader objective of this study by enabling the implementation of a graphite swelling model within Gen-Foam and its final validation against experimental data.

CRediT authorship contribution statement

S. Amirkhosravi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **A. Scolaro:** Writing – review & editing, Validation, Supervision, Software, Methodology, Conceptualization. **F. van Niekerk:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Chat-GPT (4.5) in order to check grammar and spelling. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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